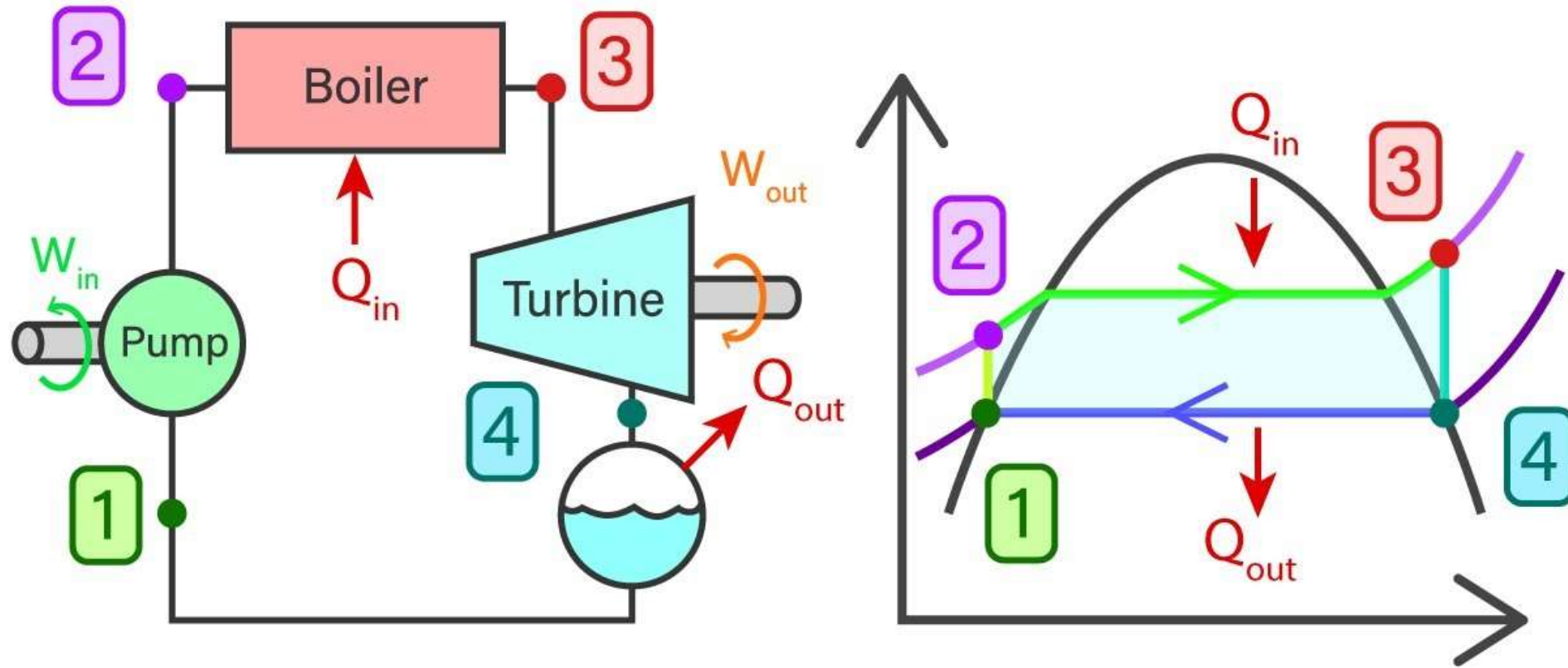


Thermal power plant

FOSSIL FUEL POWER PLANT



IDEAL RANKINE CYCLE





INTRODUCTION

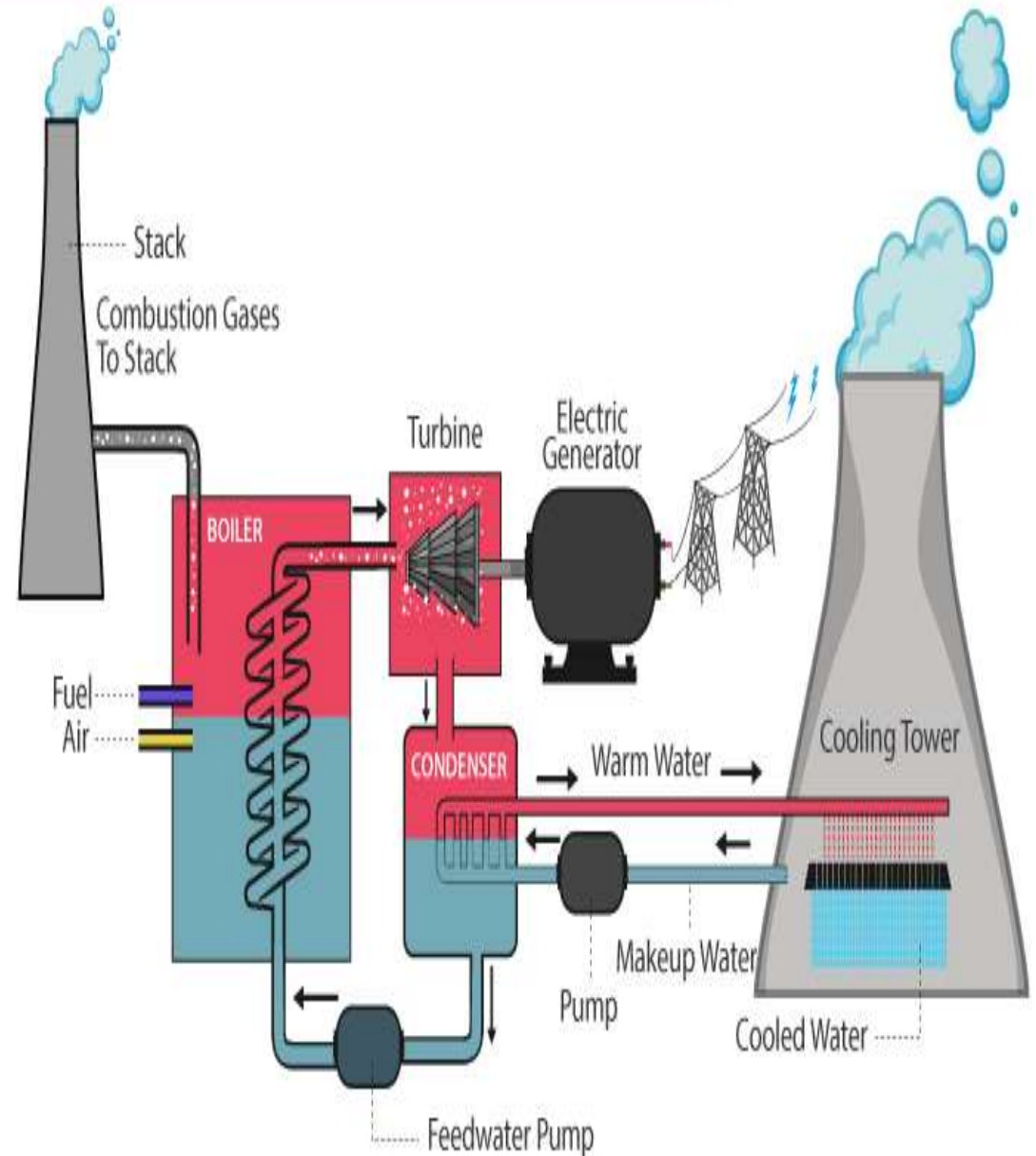
- A Thermal Power Plant converts the heat energy of coal into electrical energy. Coal is burnt in a boiler which converts water into steam. The expansion of steam in turbine produces mechanical power which drives the alternator coupled to the turbine. Thermal Power Plants contribute maximum to the generation of Power for any country.
- Thermal Power Plants constitute 75.43% of the total installed captive and non-captive power generation in India.
- In thermal generating stations coal, oil, natural gas etc. are employed as primary sources of energy.

WORKING PRINCIPLE

A thermal power plant operates based on the **Rankine Cycle**, using water and steam as the working fluid. In the boiler, water is heated by burning fuel, producing superheated steam. This steam drives a turbine connected to a three-phase synchronous generator, producing electricity. After the steam passes through the turbine, it condenses in a condenser, creating a low-pressure environment that increases efficiency. The condensate, along with makeup water, is pumped back into the boiler. A cooling water circuit cools the steam, while air and flue gases are managed by Forced Draught (FD) and Induced Draught (ID) fans.

Inside the boiler, heat exchangers like the **Economiser** preheat feed water using flue gas heat, boosting efficiency. The **Evaporator** (downcomer riser circuit) evaporates water in tubes, while the **Boiler Drum** maintains natural circulation of the steam-water mixture. Additionally, the **Superheater** superheats the steam before it goes to turbine. **Reheater, economizer and Air Preheater** further increase the efficiency.

THERMAL POWER PLANT BLOCK DIAGRAM



Main Components

- Coal Handling equipment
- Ash Handling Equipment
- Steam generator
- Reheater
- Steam Turbines
- Condenser
- Feed pump
- Economizer
- Airpreheater

Layout of Thermal Power Plant

